A1 Reproduction and Verification

Commands, environment and what the automatic checks do and do not cover

Step	What	Command (PowerShell, repository root)
1 Environment	Python 3.12.3, CPU; exact run versions in requirements-lock.txt and run.json	`pip install -r requirements-lock.txt -r requirements-timestral.txt`
2 Data (not in Git)	NQ 1-minute bars 2006–2025 (TradeStation, licensed); Timestral-5M weights in the local Hugging Face cache	`$env:FUTURES_DATA_DIR = "C:\path\to\futures-data"`
3 Train and forecast	all models, five timeframes	`python multitarget.py --tag final`
4 Tables and charts	results, rankings, timeframe map, charts, deck_data.json	`python multitarget_report.py <run> reports\multitarget_final`
5 Sample and Timestral checks	appendix A3 and A4 (read the 1-minute data / saved forecasts)	`python tools\sample_checks.py <run> reports\multitarget_final` `python tools\timestral_same_sample.py <run> reports\multitarget_final`
6 Copy, verify, build	copy deck_data.json, charts and the two JSON files to presentation\	`python verify_multitarget.py <run> reports\multitarget_final presentation` `cd presentation; npm ci; node build_deck.js`

Verified

- every point estimate in the results tables, recomputed independently from the saved forecasts
- the ranking rules, the five-model tables, the timeframe map, the data profile and the deck data
- past-to-future Timestral calibration: no calibration target after the forecasts it is used for
- unit tests (pytest) for targets, timing, splits, bootstrap and calibration

Not verified

- the raw 1-minute data and the candle building (checked separately, A3)
- the training: models were not re-fitted
- bootstrap intervals: only their order is checked
- the run used an uncommitted version of multitarget.py (hash in run.json); later changes: majority-class fix applied to saved outputs, docstrings

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A2 Selection and Statistics

How models were tuned, ranked and given uncertainty intervals

- **Tuning:** hyper-parameters (Ridge α, logistic C) and early stopping (XGBoost, networks) use the validation block (2017-08-25 → 2021-07-14) only.
- **Evidence score z:** paired validation improvement over the task's naive benchmark (squared-error reduction vs zero; accuracy gain vs majority; squared log-RV error reduction vs persistence) divided by its moving-block bootstrap standard error (20-session blocks, 2,000 replicates).
- **Revision (not blind):** the first volatility score was the QLIKE reduction (Patton, 2011). On validation a few limit-locked March 2020 candles with almost zero variance dominated it, so after a first report had printed test numbers it was replaced by the log-RV squared-error reduction.
- **Five-model tables (slides 6–8):** one model per family on each task's best validation timeframe, ranked by z. Timeframe map (slide 4): the highest-z trained model per task and timeframe.
- **Intervals:** differences (accuracy gains) use the bootstrap of paired per-row differences; ratios (R^2-type error cuts) resample model and benchmark losses together and recompute the ratio in each replicate (Künsch, 1989).
- **Majority-class fix:** the first run counted dojis as "not up"; the majority class was recomputed from non-doji training candles without refitting any model (tools/mt_fix_majority.py).

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A3 Sample Checks

Why the daily test ends on 2025-01-30, and whether incomplete candles matter

A. Daily sample end (measured, tools/sample_checks.py)

- **Cause:** all 85 sessions from 2025-01-31 to 2025-05-30 are dropped. 84 fall within 50 sessions after a session with a flagged 1-minute bar (|1-min log return| > 2%); 2025-01-31 itself is dropped because its label session (2025-02-03) contains one.
- **Flagged bars look like genuine moves, not bad prints:** e.g. 2025-02-03 (−2.7%), 2025-04-07 (−4.5%), 2025-04-09 (+2.6%). 9 of the 16 flags in 2006–2025 come right after a gap of more than an hour (weekend, holiday or daily break); the 2022 flags fall at 8:30 ET, when US macro data are released.
- **Impact:** 304 of 1002 daily test sessions (train 150, validation 115) lie in such windows, so the daily sample under-represents turbulent periods. Intraday the window is 50 candles, a negligible loss.
- **Not corrected here:** a fix (e.g. not flagging opening gaps) changes the sample, so it needs a new daily run; the saved results are kept as they are.

B. Full-length candles with missing 1-minute bars

Candle	All years	Test	Test, 8:30–15:00 CT	Test labels affected	DNN 128 error cut (%) all → complete
5 min	18.0%	0.23%	0.003%	601	39.99 → 40.00
15 min	25.8%	0.43%	0.008%	385	35.06 → 35.07
30 min	30.6%	0.65%	0.008%	279	41.72 → 41.80
1 hour	34.3%	1.01%	0.000%	201	41.97 → 42.24

- A full-length candle with 4 of its 5 minutes is accepted as complete; the count is stored (n_minutes) but not used (unit test documents it).
- Missing minutes are mostly overnight in 2006–2017, consistent with thin trading (no bar is written for a minute without trades), and almost absent in regular hours. In the test, a few sessions with long gaps coincide with quarterly contract-roll days (e.g. 2021-09-09, 2022-09-08): dataset gaps.
- Rescoring the saved test forecasts without incomplete label candles moves the rescored metrics (HAR-OLS, DNN, XGBoost; return, direction, volatility) by at most 0.4 points: no change to the conclusions.

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A4 Timestral on the Same Sample

Mistral's Timestral-5M vs the trained models on exactly the same random test candles (saved forecasts)

Candle	Sample	Return R^2: Timestral	Return R^2: DNN	Direction pp: Timestral, body	Direction pp: Timestral, close-to-close	Direction pp: DNN, body	Vol. error cut: Timestral	Vol. error cut: DNN	Corr. with next log RV: spread / current RV
5 min	20,000	−2.55%	+0.20%	−0.59	−0.66	+0.51	+12.2%	+39.6%	0.76 / 0.76
15 min	20,000	−2.23%	−0.10%	+0.01	−0.08	+1.09	−37.1%	+34.8%	0.74 / 0.84
30 min	20,000	−1.46%	+0.02%	−1.76	−1.88	+0.14	−56.2%	+41.3%	0.71 / 0.84
1 hour	20,000	−1.59%	+0.00%	−2.28	−2.36	+0.02	−74.3%	+41.8%	0.64 / 0.83
1 day	642	−1.83%	+0.17%	−1.87	−1.56	+0.00	−19.3%	+18.7%	0.75 / 0.82

- **Target mismatch is not the reason:** Timestral forecasts the next close, the study scores the candle body; scored on the close-to-close direction it does no better (columns 5–6).
- **Same rows:** the trained models keep their advantage on Timestral's own sample, so the gap is not a sampling artefact.
- **Volatility is indirect:** Timestral's volatility comes from the spread of its PRICE forecast (q10–q90), mapped to log RV by a linear calibration; the trained models forecast log RV directly from past RV. The raw spread correlates with next log RV slightly less than current RV does, and the calibration fails beyond 5 minutes.
- **Scope:** one small model (5M), frozen weights, 512-close context, one market: not a verdict on foundation models in general.
- **Proposed next experiment (not run):** aligned targets — forecast open and close of t+1 (body direction) and the quantiles of the next candle's 1-minute returns, then derive RV from Timestral's own path; compare 28M–209M on the same sample.

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A5 Data Profile

Descriptive statistics of the three targets, validation and test blocks (data_profile.csv)

Block	Candle	Rows	Sessions	UP / DOWN / doji	Return s.d. (bp)	ACF return	Mean log RV	RV rises	ACF log RV
Validation	5 min	269,167	2017-08-25 → 2021-07-14	48.8% / 47.2% / 4.0%	7.0	−0.006	-16.13	48.9%	+0.76
Validation	15 min	88,336	2017-08-25 → 2021-07-14	50.5% / 47.3% / 2.2%	12.0	+0.003	-14.81	47.9%	+0.85
Validation	30 min	44,513	2017-08-25 → 2021-07-14	51.3% / 47.1% / 1.6%	16.3	−0.001	-14.06	46.2%	+0.85
Validation	1 hour	21,642	2017-08-25 → 2021-07-14	52.5% / 46.3% / 1.2%	22.7	+0.011	-13.28	47.3%	+0.84
Validation	1 day	822	2017-08-25 → 2021-07-13	57.7% / 41.8% / 0.5%	101.5	−0.052	-9.69	45.4%	+0.84
Test	5 min	273,340	2021-07-15 → 2025-05-30	49.9% / 48.3% / 1.8%	7.7	−0.005	-15.75	48.9%	+0.75
Test	15 min	90,189	2021-07-15 → 2025-05-30	50.7% / 48.3% / 1.0%	13.3	+0.003	-14.45	47.3%	+0.84
Test	30 min	44,395	2021-07-15 → 2025-05-30	51.1% / 48.1% / 0.8%	18.5	−0.002	-13.69	46.0%	+0.84
Test	1 hour	21,507	2021-07-15 → 2025-05-30	51.5% / 48.0% / 0.5%	26.2	−0.000	-12.90	47.7%	+0.83
Test	1 day	642	2021-07-15 → 2025-01-30	54.2% / 45.8% / 0.0%	116.9	−0.001	-9.26	49.1%	+0.81

ACF = lag-1 autocorrelation (intraday: pairs within a session; daily: consecutive sessions). RV = sum of squared 1-minute log returns inside the candle.

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